CMUC Online Seminar on Algebra, Logic and Topology

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Zariski-type spectra of localic ring-like structures

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Abstract

The Zariski spectrum provides a way to associate a locale to a commutative ring. Other spectrum constructions include the Stone spectrum of a distributive lattice and the prime spectrum of an MV-algebra. In many cases the algebraic structure comes equipped with a topology which must be taken into account, as with the Gelfand spectrum of a commutative C*-algebra and the Hofmann-Lawson spectrum of a continuous lattice. We define a notion of spectrum of a (non-distributive) commutative localic semiring which includes all these examples as special cases. Furthermore, we define a quantalic spectrum which generalises the quantale of ideals of a ring and from which the aforementioned localic spectrum can be recovered. By leveraging dualisable objects in the monoidal category of suplattices, we describe an explicit construction of this spectrum under certain conditions